A-level

ENVIRONMENTAL SCIENCE

Paper 2 - Generated practice paper Set 23

Mark scheme

120 marks: Questions 1-10 (95 marks) plus one essay from Question 11 (25 marks).

Marking guidance

• Each part opens with a line in bold saying how its marks are shared out, followed by one sentence on how to mark it.
• The figure in bold at the start of a marking point is the mark that point earns. Lines indented beneath it, starting with a dash, are the detail that belongs to the same mark; a line starting **OR** is another route to it.
• **Also accept** lists other wordings and answers that earn the mark - wording never has to match these notes, only the science. **Do not credit** lists the answers that look right and earn nothing.
• Where a question asks for a fixed number of points (eg *two* reasons), credit only the first points given.
• For levels-of-response questions, determine the level from the descriptors first, then position the mark within the level using the indicative content.
• Calculations: credit correct working (method marks) even if the final answer is wrong; incorrect answers with no working score 0.

Textbook references: every question below cites the printed page numbers in R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018) from which the content is drawn, together with the AQA 7447 specification reference.

Question	Marking guidance	Marks
01.1	**1 mark each** 1 mark for each of the 3 points below. **1** A larger area supports larger populations, which are less likely to be lost to chance events or inbreeding, and it can hold species with large territories such as top predators **1** A larger area holds a wider range of habitats and microhabitats, so more species can find the conditions they need **1** A compact, rounded shape has the smallest perimeter for its area, so less of the reserve is affected by edge effects such as wind, drier conditions, spray drift and disturbance from outside **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3

Do not write outside the box

Question	Marking guidance	Marks
01.2	**1 mark each** 1 mark for each of the 2 points below. **1** It allows individuals to move between the two areas, so the populations interbreed as one larger gene pool and inbreeding is reduced **1** An area that loses a species can be recolonised from the other, and species can shift their range as the climate changes rather than being trapped in an isolated fragment **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **gene pool** → the range of alleles / genes present in a population	2
01.3	**1 mark each** 1 mark for each of the 4 points below. **1 Water levels:** sluices and bunds hold water back to create ponds, scrapes and seasonally flooded marsh, adding wetland habitats and a range of depths for different species **1 Soil nutrient levels:** cutting and removing hay rather than leaving it, and never fertilising, lowers the nutrient level so that vigorous grasses cannot out-compete the wildflowers of species-rich grassland **1 Topography:** banks, scrapes and south-facing slopes create warm and cool, wet and dry microhabitats within a small area, each suiting different species **1 Light levels:** thinning, coppicing and cutting rides and glades lets light reach the woodland floor, so ground flora and the insects that depend on it can grow **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **abiotic** → non-living / physical factors • **biodiversity** → the variety of life: habitat, species and genetic diversity	4
01.4	**1 mark each** 1 mark for each of the 3 points below. **1** A plagioclimax is a community held at an earlier stage of succession than its natural climax by continued human activity, so it never develops into woodland **1** Grazing animals eat the tree and shrub seedlings that would otherwise shade out the grassland, so succession is deflected and the open habitat survives **1** The species, stocking density and timing of grazing are chosen to suit the habitat – cattle graze less selectively and their trampling creates bare patches for seedlings, while sheep graze close and produce a short, even sward **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **plagioclimax** → a community held back from its climax by human activity • **succession** → the gradual change in the community over time **Do not credit** • 'the climax community never changes' - it is stable only while the conditions stay the same	3

Do not write outside the box

Question	Marking guidance	Marks
01.5	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Invasive non-native species such as rhododendron or Himalayan balsam out-compete and shade out native species, and have no natural controls here **1** Introduced predators or competitors, such as grey squirrels or mink, reduce native populations directly **1** Without their natural predators, herbivore numbers such as deer can exceed the carrying capacity, preventing tree regeneration and damaging the ground flora **1** Culling may also be needed to keep a population within the reserve's carrying capacity so that it does not damage the habitat the whole community depends on **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **carrying capacity** → the largest population the area can support **Do not credit** • 'the population cannot grow any more' as a definition - numbers fluctuate around the carrying capacity	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 40-49; AQA spec 3.1.3.3-3.1.3.4.	
02.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** The readings are precise but may not be accurate **Also accept** • **precise** → readings close together / fine resolution • **accurate** → close to the true value	1
02.2	**1 mark each** 1 mark for each of the 2 points below. **1** A result that differs from other results with which it was expected to be similar **1 Repeat the study / measurement:** if it recurs it is a 'real' result caused by another variable – if not it was an error and can be excluded **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **anomalous** → an outlier / a result that does not fit the pattern **Do not credit** • discarding a result only because it does not fit the expected pattern, with no reason to think it is faulty	2

Do not write outside the box

Question	Marking guidance	Marks
02.3	**1 mark each** 1 mark for each of the 2 points below. **1** To test the reliability of the method / equipment **1** To decide the sample size, number of samples, spacing or timing needed to detect the variation without collecting unnecessary data **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **reliability** → repeatable / gives consistent results when repeated **Do not credit** • 'repeat the readings to make them more accurate' - repeats improve reliability; accuracy depends on the method and the instrument	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 390-391; AQA spec 3.7.1.	
03.1	**1 mark each** 1 mark for each of the 2 points below. **1** Its reading is adjusted to match a meter known to be accurate so that the results are accurate / comparable **1** If it cannot be adjusted the percentage error should be recorded so the results can be corrected **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **accurate** → close to the true value	2
03.2	**Any 3 of 6** 1 mark each for any 3 of the 6 points below, in any order, maximum 3. **1 Droppings:** diet, gender, territory size **1 Owl pellets:** diet **1** Feeding marks (eg chewed hazel nuts): presence of species such as dormice **1 Nests / burrows:** presence and breeding **1 Tracks / footprints:** presence, movements **1** Territorial marks (scratching posts): presence and territory **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
03.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bat detectors / sonograms record the ultrasonic sounds bats produce, which differ between species **1** The recordings provide evidence of presence, abundance and activity of species that cannot easily be seen **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
03.4	**Any 3 of 4** 1 mark each for any 3 of the 4 points below, in any order, maximum 3. **1** Marked individuals mix freely with the rest of the population **1** Marks do not fall off or wear away, and marking does not affect survival **1** The population does not change during the study (no births, deaths or migration) **1** All individuals have an equal chance of being caught / animals are mobile and can be caught **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **mark-release-recapture** → estimating population size from the proportion of marked animals recaptured **Do not credit** • Points after the first three - the question asks for three, so a list of extras cannot pick up the marks.	3
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 402, 405, 407; AQA spec 3.1.2.3.5-3.1.2.3.6.	
04.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1 A:** Its mass	1
04.2	**1 mark each** 1 mark for each of the 2 points below. **1** The 24-hour rotation period means the surface is not exposed to sunlight for long, so temperature extremes / excessive heating and cooling are reduced **1** The axis is tilted relative to the orbit so the amount of insolation at each latitude changes through the year, producing seasonal variations in conditions **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **insolation** → incoming solar radiation / energy arriving from the Sun	2
04.3	**1 mark each** 1 mark for each of the 2 points below. **1** The distance from the Sun gives surface temperatures between 0 and 35 °C in most areas: too close and water would boil / evaporate, too far and it would freeze **1** Atmospheric pressure (produced by the mass of gases held by gravity) is high enough to stop the water boiling / evaporating away – at much lower pressure there would be no liquid water **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 5-7; AQA spec 3.1.1.1.	

Do not write outside the box

Question	Marking guidance	Marks
05.1	**Any 4 of 5** 1 mark each for any 4 of the 5 points below, in any order, maximum 4. **1** Copying the structural / functional adaptations of living organisms in the design of manufactured items **1** Aircraft wingtip fins copied from splayed wingtip feathers of soaring birds, reducing turbulence and fuel use **1** Humpback whale flipper tubercles copied on ship rudders and wind turbine blades (tighter turns, rotation at lower wind speeds) **1** Shark skin ridges reducing friction / bacterial adhesion (surface coatings, hospital surfaces) **1** Termite mound convection copied for natural ventilation of buildings – hollow strutted bird bones in lightweight roofs – gecko toe adhesion – burr hooks - Velcro – lotus leaf self-cleaning glass – spider silk for airbags / body armour – bat / dolphin echolocation - ultrasound scanners **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'wind turbines can generate all the time' - output depends on wind speed, and they shut down in very high winds	4
05.2	**1 mark each** 1 mark for each of the 3 points below. **1** Plants produce toxic chemicals (eg alkaloids) to defend themselves against herbivores – in controlled doses these affect human physiology **1 Examples:** morphine / codeine from poppies – quinine from cinchona bark (malaria) – aspirin from willow bark – taxol from yew (cancers) – diosgenin from Mexican yam (contraceptives, cortisone) **1 Animal examples accepted:** AZT from a marine sponge (HIV), shark liver lipids (tumours) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	3
05.3	**1 mark each** 1 mark for each of the 2 points below. **1** Only a very small proportion of species have been studied for medicinal substances so unknown species may contain valuable chemicals **1** Conserving the habitat protects all its species, including those not yet investigated (and their interdependent relationships) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2

Do not write outside the box

Question	Marking guidance	Marks
05.4	1 mark for the point below. **1** Squid giant nerve cells - nerve function / heart disease, stroke – marsupials (kangaroo, wallaby pouch) - embryo development – hippo skin secretion - sunscreen / burns – sea urchin embryos - teratogen testing – armadillo - leprosy – marine sponge proteins - transplant rejection **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 10-15; AQA spec 3.1.2.1.1-3.1.2.1.2.	
06.1	**1 mark each** 1 mark for each of the 4 points below. **1** Silver carp eat phytoplankton, bighead carp zooplankton, grass carp and tilapia vegetation, common carp bottom-feed on molluscs / insects, catfish or perch eat small fish **1** Species occupy different niches / trophic levels so they do not compete and all food sources are used **1** Bottom feeders re-suspend nutrients boosting plankton / plant growth **1** Predators control over-breeding species (tilapia) so remaining fish grow larger – total yield exceeds any single species **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4
06.2	**1 mark each** 1 mark for each of the 4 points below. **1 IMTA:** fed species (salmon, shrimp) alongside inorganic extractive species (seaweeds absorbing ammonia and phosphate) and organic extractive species (filter-feeding mussels eating waste particles / plankton) **1** Wastes of one species become nutrients for another so less pollution and extra harvests **1 Aquaponics:** aquaculture water rich in nutrients / organic matter feeds hydroponic salad crops (lettuce, herbs, watercress) **1** Plants remove the nutrients that would cause deoxygenation and cleaned water returns to the fish (tilapia, carp, catfish) – rice-fish paddies similar **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming • **deoxygenation** → loss of dissolved oxygen from the water **Do not credit** • 'energy is lost between trophic levels' on its own - say how it is lost: respiration, excretion, egestion and uneaten parts	4

Do not write outside the box

Question	Marking guidance	Marks
06.3	**1 mark each** 1 mark for each of the 2 points below. **1** Bluefin tuna do not breed well in captivity so young fish are caught from the wild and fattened in sea cages **1** Production still depends on (and removes juveniles from) the wild population – carnivores need large fish inputs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **aquaculture** → fish or shellfish farming **Do not credit** • quotas described as solving overfishing on their own - undersized fish are still caught and discarded	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 357-358; AQA spec 3.5.2.3.	
07.1	**Correct answer only** 1 mark for the correct option. No mark if more than one box is ticked. **1** B	1
07.2	**Any 4 of 6** 1 mark each for any 4 of the 6 points below, in any order, maximum 4. **1** Harvesting no faster than the MSY - resource never depleted **1** Mixed species plantations - more food choices, relationships and niches **1** Indigenous species - support indigenous insects, birds **1** Mixed age structure - varied light, temperature, wind so more species – continuous cover and yield **1** Selective logging - small clearings, less disturbance, long-term timber benefit **1** Retention of dead wood, buffer zones, corridors **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **MSY** → the largest catch that can be taken without reducing the stock **Do not credit** • Points after the first four - the question asks for four, so a list of extras cannot pick up the marks.	4
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 360-369; AQA spec 3.5.3.	

Do not write outside the box

Question	Marking guidance	Marks
08.1	**1 mark each** 1 mark for each of the 4 points below. **1** Stable tropical climate over long periods gave time for species to evolve and adapt to local abiotic and biotic factors – many rare species with small ranges **1** High light, warm temperatures and high rainfall all year allow continuous photosynthesis and growth, supporting rich food webs with no seasonal food shortage (no hibernation / migration) **1** Stratification (emergent, canopy, understorey, shrub, ground layers) and epiphytes create many niches – intense competition for light drives specialisation **1** Biotic pressures (predation, inter-species relationships) drive evolution of specialised pollination / seed dispersal relationships – plants flower and fruit at different times **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **biodiversity** → the variety of life: habitat, species and genetic diversity • **abiotic** → non-living / physical factors • **biotic** → living factors / effects of other organisms	4
08.2	**1 mark each** 1 mark for each of the 2 points below. **1** Adapted to stable conditions so they have narrow ranges of tolerance – a small change can move conditions outside the range **1** Many species are rare with small ranges / small populations **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **tolerance** → the range of conditions in which a species can survive	2
08.3	**1 mark each** 1 mark for each of the 2 points below. **1** No cold winter kills herbivores so plants defend themselves with toxic chemicals (alkaloids) that are medicinally active in controlled doses (quinine, diosgenin, curare) **1** Animals without teeth / claws (frogs, reptiles) rely on chemical defences or venoms (fer-de-lance venom - captopril) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
08.4	**1 mark each** 1 mark for each of the 2 points below. **1** Over half the Amazon's rainfall comes from transpiration within the forest – clearance reduces downwind rainfall **1** Rapid decay recycles nutrients so soils are shallow / poor – canopy and roots prevent erosion of the thin soil - clearance leads to rapid erosion and infertility **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **transpiration** → water lost as vapour from the leaves	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 52-55; AQA spec 3.1.2.3.4.	

Do not write outside the box

Question	Marking guidance	Marks
09.1	**1 mark each** 1 mark for each of the 2 points below. **1** Yield increases as inputs increase **1** But each extra unit of input produces a smaller increase in yield than the previous one (the curve flattens / law of diminishing returns) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.2	**Calculation** A correct final answer scores full marks on its own. If the answer is wrong, each correct step of working below earns its mark. **1 Second unit:** 70 - 50 = 20 units **1 Fifth unit:** 94 - 90 = 4 units **Also accept** • The correct answer however it is set out, including a different but valid method. • An answer carried forward from an earlier incorrect step (error carried forward). • An answer rounded to 2 or more significant figures, and the answer written as a fraction, decimal or percentage where the question does not specify. • The value alone - the unit (units) is already printed on the answer line. **Do not credit** • A bare wrong answer with no working.	2
09.3	**1 mark each** 1 mark for each of the 2 points below. **1** The first units of input on each area give the largest yield increases **1** Concentrating inputs means later units add very little, whereas the same inputs spread thinly all work at the steep part of the curve **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.4	**1 mark each** 1 mark for each of the 2 points below. **1** Some areas have much greater potential productivity so investment there is sensible **1** Affluent societies can afford to buy more inputs while poorer communities that would benefit most cannot afford them (economic reasons) **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
09.5	**1 mark each** 1 mark for each of the 2 points below. **1** Yield per unit area is very high because of large inputs **1** But the extra yield per unit of input is small, so efficiency (yield per unit input / energy ratio) is low **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	2
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 324-325; AQA spec 3.5.1.2.	

Do not write outside the box

Question	Marking guidance	Marks
10.1	1 mark for the point below. **1** Development that meets the needs of the present generation without compromising the ability of future generations to meet their own needs **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • **sustainable** → meeting present needs without stopping future generations meeting theirs **Do not credit** • 'sustainable means it lasts a long time' - it means use that can continue without reducing what is available in future	1
10.2	**1 mark each** 1 mark for each of the 4 points below. **1** Negative feedback resists change / returns the system towards its previous state **1** eg increased temperature increases cloud cover and albedo, or higher CO_2 increases photosynthesis **1** Positive feedback increases the change / makes it self-sustaining **1** eg melting permafrost releasing methane, declining albedo as ice melts, forest and peat fires **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • The converse of any point, ie the same comparison written the other way round. • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **positive feedback** → a change that makes itself larger / a self-reinforcing cycle • **albedo** → reflectivity / the proportion of solar radiation reflected / how much sunlight is bounced back **Do not credit** • 'ice reflects the heat back' - albedo is the reflection of incoming short-wave solar radiation, before it is absorbed	4
10.3	1 mark for the point below. **1** Many species with a complex food web so the loss of one has a small effect – agroecosystems have few species and lack detritivores, pollinators and pest predators that maintain stability **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes.	1

Do not write outside the box

Question	Marking guidance	Marks
10.4	**Levels of response** Decide the level from the descriptors first, then use the indicative content to place the mark within that level. The points below are not a checklist. **Level 3 (7-9 marks)**: Answer is comprehensive, relevant and logically structured. Shows detailed knowledge and understanding of the relevant scientific ideas and applies them accurately to the context. Evaluation/analysis is well supported by evidence and a clear judgement or conclusion is reached. **Level 2 (4-6 marks)**: Answer is mostly relevant and shows some structure. Knowledge and understanding of relevant ideas is sound but application to the context or evaluation may be incomplete or lack detail. A judgement may be made but is not fully supported. **Level 1 (1-3 marks)**: Answer is basic, with limited structure. Some relevant knowledge is shown but there are omissions or errors. Little application to the context and little or no evaluation. **0 marks**: No relevant content. **Indicative content - a guide to placing the mark, not a checklist** • **Principles of natural systems:** dynamic equilibria and negative feedback – diverse systems resistant to change – low energy-density renewable energy (solar-driven hydrological, carbon and nitrogen cycles) – low-temperature processes using enzymes – material cycles with wastes as raw materials – small number of abundant elements – biodegradable non-toxic wastes • **Human systems:** non-renewable energy, high temperatures (Haber process, incineration, manufacturing), linear resource use, dispersal of minerals and mixtures that are hard to separate, persistent toxins (PCBs, organochlorines, toxic metals in electronics) • **Applications:** renewable energy and heat pumps, bioleaching, recycling and design for re-use, industrial symbiosis, organic and integrated agriculture, retaining hedgerows and pollinators, circular economy • Human responses that override feedback (fertilisers, irrigation, pesticides) and positive feedback triggered by human activity (permafrost, methane hydrates, fires) – tipping points • **Barriers:** population size and affluence, attitudes of urban populations to distant resources, economics, need for international cooperation, existing infrastructure • Justified conclusion on how far natural principles can be applied **Also accept** • Any other correct, relevant point of the same kind - wording never has to match these notes. • Any accurate, relevant material that is not in the indicative content.	9
	• **sustainable** → meeting present needs without stopping future generations meeting theirs • **negative feedback** → a change that cancels itself out / a self-correcting cycle • **biodegradable** → broken down by decomposers **Do not credit** • 'renewables cause no pollution at all' - manufacture, construction and the land they occupy all have impacts • pests 'getting used to' a pesticide - the few naturally resistant individuals survive and pass the trait on	
	Model answer - Level 3 (8 marks)	

Do not write outside the box

Question	Marking guidance	Marks
	Natural systems have remained sustainable because they run on low energy-density renewable energy, mainly sunlight driving the hydrological, carbon and nitrogen cycles; their reactions occur at low temperatures because enzymes lower activation energies; materials cycle so the waste of one process is the raw material of the next; only a few abundant elements are used; and wastes are biodegradable and non-toxic. Negative feedback keeps them in dynamic equilibrium and high diversity makes them resistant to change.	
	Human systems contrast on every point: non-renewable fossil fuels, high-temperature processes such as the Haber process and incineration, linear use that disperses minerals and mixes materials that cannot be separated, and persistent toxins like PCBs and the metals in electronics. We also override negative feedback - when degraded soil reduces yields we add fertiliser, irrigation and pesticides - and trigger positive feedback such as permafrost melting.	
	Copying natural principles is already possible in many areas: renewable energy and heat pumps, bioleaching at low temperature, recycling and design for disassembly, industrial symbiosis, legumes and organic matter instead of manufactured nitrate, and retaining hedgerows for pest predators - the basis of the circular economy. The limits are population size and affluence, the attitude of urban consumers who never see depleted resources, cost, and the need for international cooperation. So the principles can make individual activities far more sustainable, but only a large-scale, cooperative shift in lifestyle would make the whole human system sustainable.	
	Weaker answer - Level 2 (5 marks)	
	Natural systems use renewable energy from the Sun and recycle materials, so nothing runs out and there is no waste. Humans use fossil fuels which will run out and produce waste that is toxic. Natural processes happen at low temperatures but the Haber process needs high temperatures. Humans could copy natural systems by using renewable energy like solar and wind, recycling more and using less fertiliser. This would make human activities more sustainable. However it would be expensive and people would have to change their lifestyle.	
	What makes the difference	
	• Level 2 states the contrast correctly but does not explain the mechanisms (enzymes, negative feedback, diversity, dispersal of minerals) that Level 3 uses.	
	• Level 3 applies the principles to specific technologies (bioleaching, heat pumps, industrial symbiosis, legumes); Level 2 stays at 'recycle more'.	
	• 'Evaluate the extent' - Level 3 distinguishes individual activities from the whole system in its conclusion; Level 2's evaluation is a single sentence about cost.	
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-374; AQA spec 3.6.1-3.6.3.	

Do not write outside the box

Question	Marking guidance	Marks
11.1	**Discuss how feedback mechanisms, tipping points and diversity influence the stability of natural systems, and how human activities affect them.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • Sustainability definitions – population and per capita impact – attitudes of rural and urban societies • **Negative feedback:** cloud cover and albedo, CO_2 and photosynthesis, evaporation and precipitation, density-dependent population regulation • Positive feedback driven by human activities: permafrost, acidification, albedo decline, methane hydrates, fires, cirrus clouds, decomposition • **Tipping points:** rare under negative feedback – large events (Permian methane hydrate) – self-sustaining change from human activities • **Diversity and resistance:** reefs and rainforests vs low-diversity agroecosystems lacking detritivores, pollinators and predators • Human responses that override feedback (fertilisers, irrigation, pesticides) – links to climate change chapter – justified conclusion *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.*	25
	Source: R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 370-372; AQA spec 3.6.1.	

Do not write outside the box

Question	Marking guidance	Marks
11.2	**Discuss the causes and consequences of deforestation and evaluate the extent to which forests can be exploited sustainably.** **Levels of response** **Level 5 (21-25 marks)**: Comprehensive and detailed knowledge and understanding drawn from across the specification, applied accurately to the question. Argument is coherent, well-structured and supported by a wide range of relevant, well-chosen examples and quantitative detail where appropriate. Interconnections between topics are made explicitly and a clear, justified conclusion is reached. **Level 4 (16-20 marks)**: Detailed knowledge and understanding with good application to the question. Argument is logically organised with relevant examples; some links between different areas of the specification are made. Conclusions are drawn but may be less fully justified. **Level 3 (11-15 marks)**: Sound knowledge and understanding, mostly applied to the question. Some structure and relevant examples, but coverage may be uneven or descriptive rather than analytical. Limited linking of topics. **Level 2 (6-10 marks)**: Basic knowledge and understanding with partial relevance. Answer is largely descriptive with few examples and limited structure; some errors or omissions. **Level 1 (1-5 marks)**: Limited, fragmented knowledge with little relevance to the question. Little structure; examples absent or inappropriate. **0 marks**: No relevant content. **Indicative content - students are not expected to cover all of these** • **Importance:** resources (timber, fuel, food, fibres, medicines) and ecosystem services (atmospheric regulation, carbon store, hydrological regulation, climate, soil conservation, habitat, recreation) • **Causes:** agriculture (subsistence and cash crops), mineral extraction, reservoirs, urbanisation, transport, unsustainable timber, fuel and fodder exploitation above MSY – trends by region • **Consequences:** loss of resources, biodiversity and species, fragmentation – hydrological changes – soil impacts – climate impacts (albedo, carbon, rainfall) • **Productivity vs biodiversity:** non-indigenous species, monocultures, close planting, simple age structure, clear-felling • **Sustainable management:** MSY harvest rates, mixed indigenous species, mixed age structure, selective logging, traditional coppicing / pollarding, FSC certification, community involvement, protected areas, debt-for-nature, afforestation programmes (China) – Rio Bravo case study	25
	• **Evaluation:** economics of slow-growing hardwoods, enforcement, demand from MEDCs, ITTO failures – judgement *Credit any relevant, accurate material, including points that are not listed here. Reward a clear line of argument and a supported conclusion.* **Source:** R. Genn, Environmental Science A level AQA (Insight & Perspective, 2018), pp. 50-57, 360-369, 386-387; AQA spec 3.5.3, 3.1.2.3.4.	

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